Physical Activity and Sports—Real Health Benefits: A Review with Insight into the Public Health of Sweden

Christer Malm [1], Johan Jakobsson [1,*], Andreas Isaksson [2]

- Author information
- Article notes
- Copyright and License information

PMCID: PMC6572041 PMID: 31126126

Abstract

Positive effects from sports are achieved primarily through physical activity, but secondary effects bring health benefits such as psychosocial and personal development and less alcohol consumption. Negative effects, such as the risk of failure, injuries, eating disorders, and burnout, are also apparent. Because physical activity is increasingly conducted in an organized manner, sport's role in society has become increasingly important over the years, not only for the individual but also for public health. In this paper, we intend to describe sport's physiological and psychosocial health benefits, stemming both from physical activity and from sport participation per se. This narrative review summarizes research and presents health-related data from Swedish authorities. It is discussed that our daily lives are becoming less physically active, while organized exercise and training increases. Average energy intake is increasing, creating an energy surplus, and thus, we are seeing an increasing number of people who are overweight, which is a strong contributor to health problems. Physical activity and exercise have significant positive effects in preventing or alleviating mental illness, including depressive symptoms and anxiety- or stress-related disease. In conclusion, sports can be evolving, if personal capacities, social situation, and biological and psychological maturation are taken into account. Evidence suggests a dose–response relationship such that being active, even to a modest level, is superior to being inactive or sedentary. Recommendations for healthy sports are summarized.

Keywords: youth, adolescent, elderly, quality of life, relative age effect, exercise, strength and conditioning

1. Introduction

Sport is a double-edged sword regarding effects on health. Positive effects are achieved primarily through physical activity, which is the main part of most sports. Many secondary effects of sport also bring health benefits, such as psychosocial development of both young [1] and old [2], personal development [3], later onset, and less consumption of alcohol [4,5]. Finally, those who play sports have a higher level of physical activity later in life [6], and through sport, knowledge of nutrition, exercise, and health can be developed [7]. Negative effects include the risk of failure leading to poor mental health [8,9], risk of injury [10,11], eating disorders [12], burnout [13], and exercise-induced gastrointestinal tract discomfort [14]. In sport, there are unfortunately also reports of physical and psychological abuse [15]. Negative aspects are more common in elite-level sports, where there is a fine balance

between maximum performance and negative health. A somewhat unexpected effect of sport participation is that people submitting to planned training in some cases perform less physical activity compared to those who are exercising without a set schedule. One explanation can be a reduced spontaneous physical activity in the latter group [16]. Because physical activity is increasingly executed in an organized manner [17,18,19], sport's role in society has become increasingly important over the years, not only for the individual but also for public health.

In this paper, we describe the health effects of sport from a physiological and psychological perspective, related both to physical activity and added values of sport per se. Initially, brief definitions of various concepts related to physical activity and health are given. This is then followed by: (1) A brief description of how physical activity and training affect our body from a physiological perspective; (2) a report on the health effects of physical activity and training; and (3) sport's specific influences on the various dimensions of health. We chose to discuss the subject from an age-related perspective, separating children/adolescents, adults, and the elderly, as well as separating for sex in each age group.

2. Definitions of Physical Activity, Exercise, Training, Sport, and Health

Definitions and terms are based on "Physical activity in the prevention and treatment of disease" (FYSS, www.fyss.se [Swedish] [20]), World Health Organization (WHO) [21] and the US Department of Human Services [22]. The definition of physical activity in FYSS is: "Physical activity is defined purely physiologically, as all body movement that increases energy use beyond resting levels". Health is defined according to the World Health Organization (WHO) as: "[…] a state of complete physical, mental and social wellbeing and not merely the absence of disease or infirmity" [21].

Physical activity can occur spontaneously (leisure/work/transport) or organized and be divided according to purpose: Physical exercise is aimed primarily at improving health and physical capacity. Physical training is aimed primarily at increasing the individual's maximum physical capacity and performance [23]. Physical inactivity is described as the absence of body movement, when energy consumption approximates resting levels. People who do not meet recommendations for physical activity are considered physically inactive and are sometimes called "sedentary". Sport can be organized by age, sex, level of ambition, weight or other groupings [24]. Sport can also be spontaneous [7,17] and defined as a subset of exercises undertaken individually or as a part of a team, where participants have a defined goal [7]. General recommendations for physical activity are found in Table 1, not considering everyday activities. One can meet the daily recommendations for physical activity by brief, high-intensity exercise, and remaining physically inactive for the rest of the day, thereby creating a "polarization" of physical activity: Having a high dose of conscious physical training, despite having a low energy expenditure in normal life due to high volumes of sedentary time. Polarization of physical activity may lead to increased risk of poor health despite meeting the recommendations for physical activity [25,26,27]. During most of our lives, energy expenditure is greater in normal daily life than in

sport, physical training, and exercise, with the exceptions of children and the elderly, where planned physical activity is more important [28].

3. Aerobic and Muscle-Strengthening Physical Activity

Physical activity is categorized according to FYSS as: (1) Aerobic physical activity and (2) muscle-strengthening physical activity. Physical activity in everyday life and exercise training is mainly an aerobic activity, where a majority of energy production occurs via oxygen-dependent pathways. Aerobic physical activity is the type of activity typically associated with stamina, fitness, and the biggest health benefits [29,30,31]. Muscle-strengthening physical activity is referred to in everyday language as “strength training” or “resistance training” and is a form of physical exercise/training that is primarily intended to maintain or improve various forms of muscle strength and increase or maintain muscle mass [32]. Sometimes, another category is defined: Muscle-enhancing physical activity, important for maintenance or improvement of coordination and balance, especially in the elderly [33]. According to these definitions, muscle-strengthening activities primarily involve the body’s anaerobic (without oxygen) energy systems, proportionally more as intensity increases.

Exercise intensity can be expressed in absolute or relative terms. Absolute intensity means the physical work (for example; Watts [W], kg, or metabolic equivalent [MET]), while relative intensity is measured against the person’s maximum capacity or physiology (for example; percentage of maximum heart rate (%HR), rate of perceived exhaustion (RPE), $W \cdot kg^{-1}$ or relative oxygen uptake in $L \cdot min^{-1} \cdot kg^{-1}$ (VO_2)). In terms of recommendations to the public, as in Table 1, the intensity is often described in subjective terms (“makes you breathe harder” for moderate intensity, and “makes you puff and pant” for vigorous intensity) [27]. While objective criteria such as heart rate and accelerometry will capture the intensity of activity, they may not distinguish between different types of physical activity behaviors [34]. FYSS defines low intensity as 20%–39% of VO_2max, <40 %HR, 1.5–2.9 METs; moderate intensity as 40%–59% of VO_2max, 60–74 %HR, 3.0–5.9 METs, and vigorous intensity as 60%–89% of VO_2max, 75–94 %HR, 6.0–8.9 METs. Absolute intensity, however, can vary greatly between individuals where a patient with heart disease may have a maximal capacity of <3 MET, and an elite athlete >20 MET [35].

4. How does the Body Adapt to Physical Activity and Training?

Adaption to physical activity and training is a complex physiological process, but may, in the context of this paper, be simplified by a fundamental basic principle:” The general adaptation syndrome (GAS)” [36,37,38]. This principle assumes that physical activity disturbs the body’s physiological balance, which the body then seeks to restore, all in a dose-related response relationship. The overload principle states that if exercise intensity is too low, overload is not reached to induce desired physiological adaptations, whereas an intensity too high will result in fatigue and possibly overtraining. Thus, for adaptation to occur, greater than normal stress must be induced,

interspersed with sufficient recovery periods for restoration of physiological balance [39]. During and immediately after physical exercise/training, functions of affected tissues and systems are impaired, manifested as temporarily decreased performance. You feel tired. In order to gradually improve performance capacity, repeated cycles of adequate overload and recovery are required [40]. In practice, positive effects can be seen after a relatively short period of a few weeks, but more substantial improvements if the training is maintained for a longer period.

As a rule of thumb, it is assumed that all people can adapt to physical activity and exercise, but the degree of adaptation depends on many factors, including age, heredity, the environment, and diet [41,42,43,44]. The hereditary factor (genetics) may be the most critical for adaptation [45]. The degree of adaptation also depends on how the person in question trained previously; a well-trained athlete usually does not have the same relative improvement as an untrained one. Even if training is thought to be specific to mode, intensity, and duration, there are some overlaps. For example, it has been found that strength training in some individuals contributes to a relatively large positive impact on health and endurance, effects previously associated primarily with aerobic exercise [46,47]. The overload principle may, if applied too vigorously in relation to a person's individual adaptation ability, have detrimental effects, including reduced performance, injury, overtraining, and disease [10]. Training is a commodity that must be renewed; otherwise, you gradually lose achieved performance improvements [48], although some capacities, such as muscle memory, seem to persist for life [49].

General recommendations for health may be stated, but individual predispositions make general training schedules for specific performance effects unpredictable. All exercise training should be adjusted to individual purposes, goals, and circumstances.

5. Health Effects of Physical Activity and Training

Human biology requires a certain amount of physical activity to maintain good health and wellbeing. Biological adaption to life with less physical activity would take many generations. People living today have, more or less, the same requirements for physical activity as 40,000 years ago [50,51]. For an average man with a body weight of 70 kg, this corresponds to about 19 km daily walking in addition to everyday physical activity [52]. For most people, daily physical activity decreases, while planned, conscious exercise and training increases [19,53]. Unfortunately, average daily energy intake is increasing more than daily energy output, creating an energy surplus. This is one reason for the increasing number of overweight people, and a strong contributor to many health problems [54]. More sedentary living (not reaching recommended level of physical activity), combined with increased energy intake, impairs both physical and mental capabilities and increases the risk of disease. Despite this, Swedes (as an example) seemed to be as physically active and stressed but had better general health in 2015, compared to 2004 (Figure 1). Compared to 2004–2007, the Swedish population in 2012–2015 reported better overall health (more county-dots are blue) and less fatigue (smaller county-dots) with similar level of physical activity (~65% indicated at least 30 min daily physical activity) and stress (~13% were stressed).